

ChessMate: Intention Disclosure to Support Sense of Agency in Human-Computer Collaboration

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Figure 1: ChessMate in-use overview. (a) Physical setup: the human collaborates with robot-controlled White pieces against a computer-controlled Black opponent. (b) On each White turn, a piece proposes a move and waits for the user's approval or rejection before execution. (c) Example utterances illustrating the four disclosure conditions: No (move only), Low (brief local justification), High (multi-step and longer-term rationale), and Misleading (spoken recommendation intentionally mismatches the executed move).

Abstract

As AI-driven systems evolve from instruction-following tools to autonomous collaborators, users may experience reduced sense of agency (SoA), the subjective perception of control, when reciprocal intent alignment is limited. This study tests whether intent disclosure, defined as communicating an intelligent system's action goals and decision rationales, increases humans' SoA in human-computer collaboration. We developed ChessMate, a collaborative chess system in which a human player teams with AI-assisted white pieces against computer-controlled black pieces. At each turn, chess-piece robots recommend a move under four conditions: No, Low, High, and misleading disclosure, and the human player can choose to approve the move or not. In a within-subjects study ($n = 16$), high disclosure significantly increased SoA relative to no disclosure,

while misleading disclosure did not reduce SoA and yielded descriptively higher scores. Qualitative findings suggest that cognitive engagement underlies these effects. Richer disclosures may support collaborative SoA in high-autonomy systems.

CCS Concepts

• Human-centered computing → Empirical studies in interaction design.

Keywords

Sense of agency, Intention disclosure, Human-Computer collaboration, Decision-making

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1 Introduction

With advances in artificial intelligence, robots are changing from tools that simply follow instructions to intelligent partners equipped with contextual awareness and autonomous decision-making capabilities. However, the rise of automation introduces a paradox: the more capable robots become, the less Sense of Agency (SoA) users experience [12]. SoA refers to the subjective perception of control, namely the extent to which users feel that “I am the one doing this; my intention is driving the outcome” when interacting with a system [1]. Reduced SoA can cause users to become overly reliant on automation. For example, when drivers lose SoA in automated vehicles, they may fail to stay sufficiently engaged in monitoring the road [10]. Conversely, when users cannot understand or override a robot’s actions, such as when an autonomous driving system suddenly takes over steering while the human driver remains liable for any resulting accident, their trust in the system deteriorates and they may attempt to bypass or abandon it [3]. This tension raises a central question: how can we maintain SoA in highly autonomous human-robot collaboration systems?

The fundamental challenge in human–computer collaboration lies in the mutual lack of understanding between humans and computer [12]. However, existing research aimed at enhancing SoA has predominantly focused on equipping robots to infer human intentions [8, 9], an approach that only addresses one side of the interactive dynamic. Even if the computer understands the human, the human cannot experience agency if the robot’s behavior remains inexplicable. We argue that intention disclosure is essential to target this gap. Unlike trajectory visualizations [11] or simple status cues [7], intention disclosure explains what the system intends to do and the information on which its decisions are based. Such explanations help users construct accurate causal mental models and regain a cognitively grounded sense of agency.

To investigate how intention disclosure shapes SoA, we consider chess an ideal experimental domain because it elicits sustained strategic reasoning in a highly controlled environment [6], allowing subtle variations in SoA to be reliably captured. We developed the intelligent chess system ChessMate, where participants collaborate with a set of white chess-piece robots against a computer-controlled black side (Figure 1.a). Each robot analyzes the board, proposes a move, and executes it upon user approval (Figure 1.b). This setup supports clear observation of SoA: players experience high SoA when they integrate the robots’ suggestions into their own reasoning, and low SoA when they follow suggestions without reflection. We manipulated four intention disclosure conditions (No, Low, High, and Misleading Disclosure) (Figure 1.c) and collected subjective SoA scores and interview results to examine how different forms of disclosure modulate agency in collaborative decision-making and to inform the design of highly autonomous systems.

2 Prototype

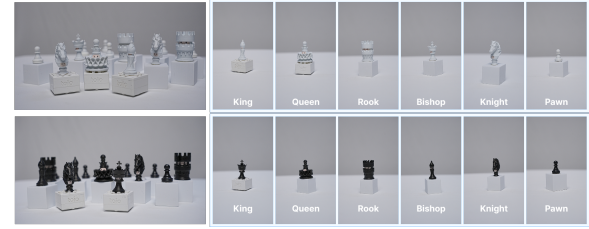


Figure 2: Chess-piece appearance design. 3D-printed shells mounted on Toio Core Cubes provide distinct silhouettes for each piece type (king–pawn) and color, enabling quick at-a-glance identification during play.

We developed ChessMate, an intelligent chess-piece system composed of multiple robotic pieces capable of parsing board states, generating move recommendations, and executing autonomous movements. Each piece uses a Sony Toio Core Cube as its mobile base, with an Arduino module managing local commands and hardware interaction, and a custom 3D-printed shell that makes different piece types clearly distinguishable (Figure 2). Unity serves as the central control hub, while the WonderEcho module supports voice interaction by integrating a microphone, speaker, and neural-network-based speech processor for speech recognition and real-time audio playback. The structure of each chess piece is shown in Figure 3.

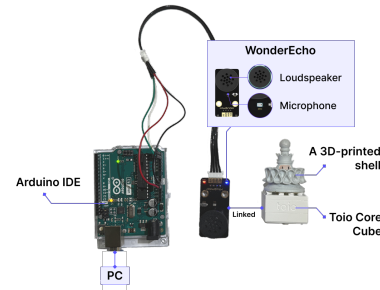


Figure 3: Hardware and control pipeline for each chess-piece robot. Unity on a PC maintains the game state, queries the LLM for a recommended move and explanation, and sends commands to an Arduino controller. Arduino triggers WonderEcho for speech playback and controls the Toio cube for motion; the 3D-printed shell provides the piece form factor.

The system incorporates an intention-generation agent powered by a large language model (LLM). At each turn, Unity encodes the current board state into a structured input for the model, which returns both a recommended move for the white piece and a corresponding natural-language explanation. The generated text is then converted to speech locally and played through WonderEcho, enabling real-time, dynamic intention disclosure.

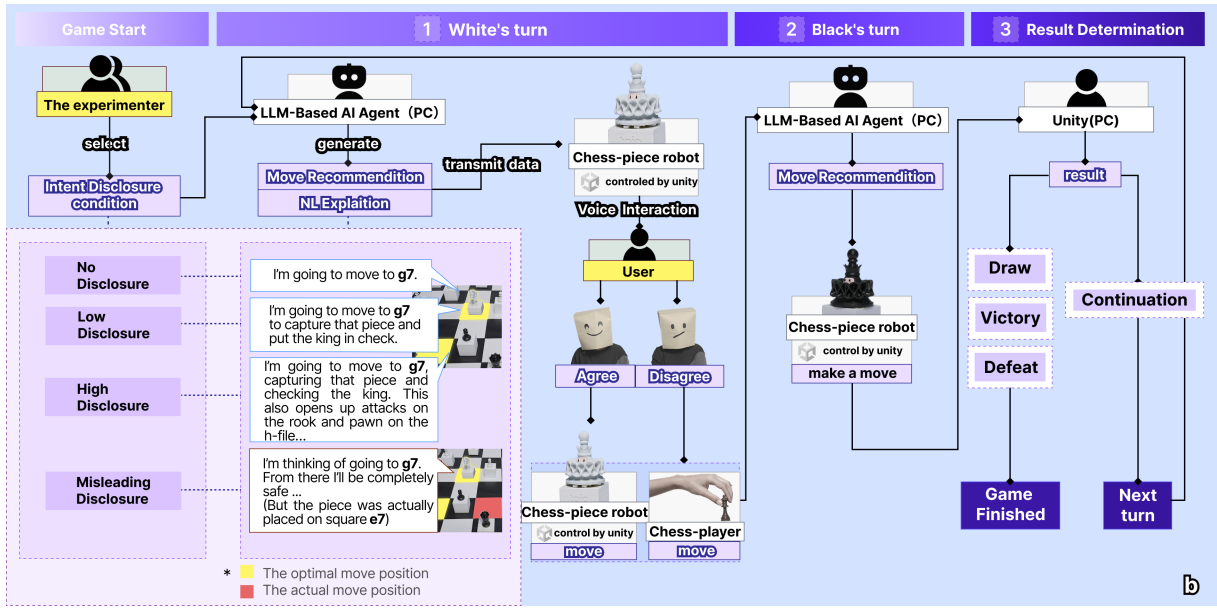


Figure 4: ChessMate disclosure manipulation and interaction loop. (a) Representative dialogue examples for the four disclosure conditions. (b) Turn-level loop: the experimenter selects a disclosure condition → on White's turn the system generates a recommendation and plays the disclosure → the user responds "agree/disagree" → if agree, the robot executes the move; if disagree, the user moves manually → on Black's turn the system moves autonomously → the game state is updated and the loop continues until checkmate or draw.

3 Interaction Design

The prototype positions the user and ChessMate as collaborative White players competing against a computer-controlled Black side. Participants were instructed that the task goal was to complete each short game under the given condition, focusing on the collaboration experience and perceived control rather than performance optimization or chess learning outcomes. To ensure experimental control and consistency, each participant is randomly assigned one of four predefined endgame configurations. In every turn, the LLM agent generates speech corresponding to the assigned intention disclosure condition, providing real-time explanations of the robot's intended actions. Jourard's four dimensions of disclosure (breadth, depth, duration, honesty) [5] are adapted to structure the robot's intention disclosure across four conditions (A to D), ranging from no explanation to misleading explanation (see Figure 4.a).

- **No Disclosure:** The robot only states the intended move, simulating action-only behavior without any explanation.
- **Low Disclosure:** The robot gives a brief tactical justification restricted to the current move and nearby local tactics, with shallow and easily understandable reasoning.
- **High Disclosure:** The robot provides multi-step reasoning and long-term strategic goals, extending both the breadth and depth of explanation to anticipated future board developments.
- **Misleading Disclosure:** The robot offers seemingly reasonable moves and justifications, but the executed move contradicts its verbal recommendation.

Each round follows a structured cycle, as shown in Figure 4.b. The interaction is organized around a fixed decision point on every White turn: the system first presents a move recommendation together with a disclosure utterance, and only executes any piece motion after the user's response is recognized. Before the experiment begins, the experimenter selects the corresponding disclosure condition in the system based on the group to which the participant is randomly assigned, and participants start after observing the current chessboard situation. During White's turn, Unity queries the large language model agent for strategy analysis, recommended moves, and corresponding explanatory text; Arduino then triggers WonderEcho to play the disclosure audio. Participants respond in natural language; WonderEcho recognizes "agree" or "disagree" in real time and Arduino sends the recognition result back to Unity. If the participant accepts the suggestion, Unity calls the Toio motion API to move the corresponding chess piece along the planned path; if the participant rejects it, the piece remains stationary and the participant manually completes the move. During Black's turn, the system independently selects and executes its move. Importantly, across conditions the timing and decision point are held constant (with voice style matched); only the granularity and truthfulness of the spoken content varies (e.g., brief move-local rationale in Low vs multi-step and longer-term goals in High, and an intentional mismatch between spoken recommendation and executed move in Misleading). The cycle continues alternately until the game ends in checkmate or a draw.

4 User Study

4.1 Experiment Design

We conducted a within-subjects study to examine whether, and through which conditions of intention disclosure, a robot can enhance users' SoA in collaborative decision-making. We proposed a hypothesis: Compared with No Disclosure, Low and High Disclosure increase SoA, while Misleading Disclosure reduces SoA.

$N = 16$ adult participants (mean age = 27; 7 male, 9 female) were recruited, all novice chess players. After a brief introduction to the robot's functions, interaction flow and the four conditions, each participant completed four short games, one under each disclosure condition. The conditions were matched in difficulty, voice style and appearance; only the detail and truthfulness of the speech content varied. Condition order was randomized across participants. At each turn, participants chose whether to accept the robot's recommendation.

Table 1: Subjective SoA Questionnaire (7-Point Likert Scale)

Item	Statement	Scale
Q1	I feel that the chess robot's moves largely match my intentions, as if it is executing my decisions.	1 = strongly disagree, 7 = strongly agree
Q2	I feel that I am in control of the chess robot's moves.	
Q3	I feel that I am the leader of each move.	
Q4	I feel that the chess robot is influencing or steering my decisions.	1 = strongly agree, 7 = strongly disagree
Q5	I feel that the chess robot is taking the lead in carrying out the actual moves.	
Q6	I feel that the chess robot has its own autonomous intentions and decisions.	

4.2 Evaluation

We collected both objective and subjective measures. Subjective SoA was assessed using a 6-item, 7-point Likert questionnaire based on Fribourg's framework [4] (Table 1), completed once per condition by each participant. For positively worded items (Q1–Q3), responses ranged from 1 (strongly disagree) to 7 (strongly agree). For negatively worded items (Q4–Q6), we used a reversed response scale (1 = strongly agree, 7 = strongly disagree) so that higher numeric values consistently indicate higher SoA across all items. We then averaged the six item scores to compute a composite subjective SoA index, where higher scores indicate a stronger sense of agency.

After the games, we conducted approximately 10-minute semi-structured interviews with each participant on perceived control and experiences of being led or manipulated. With consent, all interviews were audio-recorded for subsequent qualitative analysis.

5 Results

5.1 Quantitative Results

First, descriptive statistics were computed for SoA scores across the four disclosure conditions (Figure 5). Low Disclosure ($M = 4.19$,

$SD = 1.41$) and High Disclosure ($M = 4.26$, $SD = 1.19$) yielded higher SoA scores than No Disclosure ($M = 3.39$, $SD = 1.22$). Notably, SoA under Misleading Disclosure ($M = 3.81$, $SD = 1.17$) also exceeded the No Disclosure condition, despite the incorrect explanations.

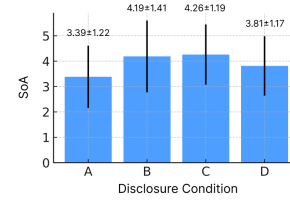


Figure 5: Figure 5: Mean composite SoA ratings (±SD) across disclosure conditions in the within-subjects study (n=16). A=No, B=Low, C=High, D=Misleading.

We then conducted inferential analyses to examine whether SoA differed systematically by disclosure level. Mauchly's test indicated that the sphericity assumption was satisfied, $W = 0.76$, $p = .59$. A repeated-measures ANOVA revealed a marginal main effect of disclosure condition, $F(3, 45) = 2.53$, $p = .069$, $\eta_p^2 = .14$, suggesting moderate within-subject variability attributable to disclosure conditions.

To probe our hypotheses, we performed planned paired-sample t -tests comparing each disclosure condition with No Disclosure. Consistent with the descriptive results, High Disclosure significantly increased SoA relative to No Disclosure condition, $t(15) = 3.40$, $p = .004$. Low Disclosure produced a positive but non-significant trend, $t(15) = 2.09$, $p = .055$. Misleading Disclosure did not significantly differ from No Disclosure condition, $t(15) = 1.19$, $p = .252$, although its descriptive mean was higher.

These findings partially support our hypothesis: correct and detailed explanations significantly increased SoA compared with no disclosure, whereas misleading disclosure did not reduce SoA as expected. Instead, it yielded a numerically higher mean than the no-disclosure condition, although this difference was not statistically significant. We therefore treat this pattern as exploratory and interpret it in light of the interview findings.

5.2 Qualitative Analysis

The interviews with 16 participants (P1–16) illuminate a common mechanism through which intention disclosure shapes SoA. Drawing on the interviews, we observed that richer intention disclosure (e.g., stating goals, rationales, and forward-looking considerations) more often led participants to engage in deeper deliberation—such as checking the reasoning, exploring variations, comparing alternatives, and reflecting before acting. This suggests that richer intention disclosure fosters higher **cognitive engagement**. Notably, these accounts of heightened engagement frequently co-occurred with participants' statements of being in charge" or "controlling the decision," aligning with a stronger sense of **SoA** during play. P6 remarked, "It sounded very certain but gave few details, so I just followed its move," describing decisions that did not feel self-initiated. Under High Disclosure, detailed explanations encouraged users to actively evaluate the robot's reasoning and make more deliberate choices. P5 noted, "Its strategy was good and it provided

me with new ideas.” P15 similarly stated, “If the explanation is too detailed but doesn’t match my thoughts, I will just follow my own plan,” indicating that richer disclosure can still support autonomous decision-making. Under Misleading Disclosure, inconsistencies triggered even stronger vigilance and independent thinking, clarifying why Misleading Disclosure produced elevated SoA in the descriptive statistics. P3 reported, “In the fourth type I can detect the trap.” P8 explained, “I always assume it’s wrong, so I don’t listen at all.” P12 added, “Under the deceptive condition, I think more and don’t follow blindly.” Taken together, these accounts suggest that higher SoA may emerge when intention disclosure elicits deeper cognitive engagement rather than when it simply increases the amount or accuracy of information.

Table 2: Paired-sample T-test Results for SoA across Intention Disclosure Conditions

Comparison	Mean diff. (I–J)	<i>t</i> (15)	<i>p</i> (2-tailed)	Cohen’s <i>d</i>
B – A	0.80	2.09	.055	0.521
C – A	0.88	3.40	.004**	0.851
D – A	0.43	1.19	.252	0.298

6 Conclusion

The study investigates how different conditions of intention disclosure influence users’ sense of agency in a collaborative chess task. We designed a robotic chess-piece prototype that explains its planned moves with four conditions of disclosure: No Disclosure condition, Low Disclosure condition, High Disclosure condition, and Misleading Disclosure condition. We combined SoA ratings and post-hoc interviews to probe how users experience control and collaboration with a robot partner. Our results show that providing High disclosure increased users’ SoA compared with offering No Disclosure. Although Misleading Disclosure did not significantly differ from No Disclosure in SoA ratings, it produced numerically higher scores rather than the expected decrease. The underlying mechanism behind these results, inferred from our post-hoc interviews, is that intention disclosure elicits users’ cognitive engagement, prompting them to actively think and make judgments, thereby enhancing their SoA. We conceptualize intention disclosure as an interaction-level cognitive trigger [2]: rather than merely increasing the amount of information, it increases the evaluability and questionability of the robot’s advice at the moment of decision, making users more likely to interrupt heuristic compliance and engage in analytic deliberation.

Although this work focuses on a chess-based collaboration setting, the underlying mechanisms we identify are relevant to other forms of human–computer collaboration and suggest two promising design implications. First, in high-risk systems where humans share responsibility for the outcomes, incorporating explicit intention disclosure is essential. In domains such as medical diagnosis, autonomous driving takeovers, or high-stakes financial decisions, system should present sufficient and intelligible justification before acting, thereby prompting users to engage in joint deliberation and helping ensure decision quality. Second, in low-risk contexts, systems may enhance users’ SoA by inviting them to participate

in error correction or trade-off evaluation. This approach is particularly relevant in educational and rehabilitation settings. For example, an educational system may intentionally reveal an obviously incorrect line of reasoning to elicit users’ corrective thinking, thereby stimulating their learning agency and engagement. These implications offer insights into selecting appropriate forms of intention disclosure when designing highly autonomous systems. Future work includes expanding the sample size to strengthen the statistical reliability of our findings, and examining the generalizability of intention disclosure across a broader range of human–computer collaboration contexts, such as education, healthcare, and manufacturing.

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